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Original Study

Palliative Rehabilitation Improves Health Care Utilization and Function in Frail Older Adults with Chronic Lung Diseases

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ABSTRACT

Objectives: The *I*ntegrated *C*are for *A*dvanced *RE*spiratory Disorders (ICARE) service is a stay-in, post-acute care program for hospitalized patients with chronic, nonmalignant lung diseases. It provides palliative rehabilitation—a novel model integrating functional rehabilitation with early palliative care. We compare reduction of health care utilization among ICARE participants vs matched controls receiving usual care.
Design: This is a prospective, propensity score–matched study. Primary outcomes were reduction in hospital readmissions and length of stay and emergency department (ED) visits, comparing the period 6 months before and after ICARE, or 6 months before and after hospital discharge (for controls). Secondary outcomes included 6-minute walking distance (6MWD) and Modified Barthel Index (MBI).
Setting: Participants were matched 1:1 to controls by age, respiratory diagnosis, socioeconomic strata, index hospitalization length of stay, frailty, and recent admissions into intensive care unit or noninvasive ventilation units.
Methods: Multidisciplinary interventions focused on symptom relief, functional rehabilitation, targeted comorbidity management, and postdischarge care coordination.
Results: One hundred pairs of patients were matched. Participants were older adults (mean age 73.9 ± 8.2 years) with prolonged index hospitalization (median 12.0 days; interquartile range 7-18). Overall, 57% had high Hospital Frailty Risk Scores and 71% had overlapping respiratory diagnoses, the most common commonest being COPD (89%), followed by interstitial lung disease (54%) and bronchiectasis (28%). Small reductions in health care utilization were observed among controls. ICARE was associated with a further 9.1 ± 19.9 days' reduction in hospitalization length of stay ($P < .001$), 0.8 ± 1.9 lesser admission ($P < .001$), and 0.6 ± 2.2 fewer ED visits ($P < .02$). Participants with longest index hospitalization were observed to have greatest reduction in length of stay. 6MWD and MBI scores improved by 41.0 ± 60.2 m and 12.3 ± 11.6 points, respectively (both $P < .001$). Greater improvement was observed in patients with lower baseline 6MWD and MBI scores. Prescription of slow-release opioids rose from 9% to 49%. Treatment for anxiety and depression rose from 5% to 19%.
Conclusions and Implications: Integrating palliative care with postexacerbation functional rehabilitation was associated with short-term reduction in health care utilization, improved functional capacity, and increased treatment of dyspnea, anxiety, and depression.

Chronic lung diseases are leading causes of mortality and disability.[1–3] By 2030, chronic obstructive pulmonary disease (COPD) will be the third-leading cause of mortality and seventh-leading cause of disability worldwide.[1] Bronchiectasis affects between 67 and 566.1 per 100,000 inhabitants in Europe and North America,[2] whereas interstitial lung diseases was the 40th leading cause for global life years lost in 2013.[3] These patients with chronic lung diseases suffer a vicious cycle of hospitalizations, skeletal muscle dysfunction, incremental incapacitation, social isolation, and psychosocial distress.[4,5]

The authors declare no conflicts of interest.

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Frailer and older persons are particularly vulnerable. Despite strong evidence of benefit, their access to both pulmonary rehabilitation (PR) and palliative care remain poor.

Since 2008, the American Thoracic Society (ATS) has advocated for palliative care to be available for symptomatic patients at any point of their progressive illnesses.[6] However, most were not referred till death became imminent.[7,8] In recent years, breathlessness services (BS) demonstrated that early palliative care improved physical and affective symptoms, quality of life (QoL) and disease self-mastery.[9,10] BS delivered face-to-face multidisciplinary and early palliative interventions at home or in the clinic, in combination with intermittent telephonic consultations. Despite improved outcomes, these services are scarce relative to mounting demand. Most patients experience distressing dyspnea underaddressed by care providers.[11] In a cross-sectional survey of 2563 patients with COPD across 5 European countries, 40% to 60% reported moderate to severe dyspnea despite treatment.[12]

Similarly, early PR improves exercise capacity and QoL, along with reductions in readmissions and mortality.[13,14] Yet PR is globally underutilized owing to limitations in availability, transport, and funding structures.[15,16] Additionally, constraints in bed capacities pressure clinicians toward rapid discharges, which disrupt care continuity and neglect the needs of frailer, older persons who require longer time to recover from acute exacerbations. This inadvertently contributes to preventable readmissions that are potentially amenable to targeted post-acute care interventions.[17]

We hypothesize that an integrated program at a dedicated unit that holistically delivers early palliative care alongside post-exacerbation functional rehabilitation, will reduce short-to medium-term health care utilization.

We describe the design and evaluation of the *I*ntegrated *C*are for *A*dvanced *RE*spiratory Disorders (ICARE) service—a novel stay-in, postexacerbation palliative rehabilitation program for chronic nonmalignant lung conditions. Interventions focused on early symptom palliation, targeted treatment of multimorbidities, patient empowerment, functional rehabilitation, and postdischarge care coordination, with the aim of alleviating total dyspnea.[18] Vertical and horizontal health care service integration were also enhanced across tertiary, rehabilitative, and community-based facilities.

This prospective, propensity score−matched (PSM) observational study aimed to compare reduction in health care utilization of ICARE participants against PSM controls who received usual care. Secondary outcomes included pre-post intervention improvement in physical function.

Ethics Approval

Approval was obtained from the National Healthcare Group Domain Specific Review Board (DSRB 2020/01148).

Methods

ICARE is an 8-bed dedicated unit within a rehabilitation hospital in Singapore. It worked in close partnership with a neighboring 1800-bed university-affiliated tertiary hospital. Between June 1, 2017, and May 31, 2020, patients were screened for eligibility during their acute hospitalizations. Consecutive patients were approached for enrolment. Eligible participants were hospitalized patients with a diagnosis of chronic lung disease. Those aged $<$18 years, with severe cognitive impairment, bedbound, or diagnosed with active cancers were excluded. Participants were accepted into ICARE once clinically stable. Patients with unplanned discharges within 7 days of enrolment and those who died within the follow-up period were excluded from analysis. For patients with $\geq$2 admissions into ICARE, only results of their first ICARE admissions were analyzed.

PSM controls were derived from a historical cohort admitted between January 1, 2016, and May 31, 2017 (before launch of ICARE), as well as from another university-affiliated tertiary hospital under the same health care cluster but in a different district, whose patients have no access to ICARE (admitted between January 1, 2016, and October 31, 2019). A digital search was performed on the health care cluster's electronic database to identify suitable controls admitted during the designated period, by applying similar inclusion and exclusion criteria outlined above.

Intervention

ICARE was designed after comprehensive literature reviews, as well as expert and stake-holder consultations involving palliative, rehabilitative, and respiratory medicine health care professionals. In contrast to conventional models of care, which had been described as fragmented, single-organ specific, and heavily reliant on acute care, ICARE adopted key elements in the conceptual framework of integrated care proposed by Valentijn et al.[19] There are 3 distinct levels of integration (see Supplementary Figure 1):

1. *Cross-specialty integration.* Evidence-based pharmacologic and nonpharmacologic interventions spanning palliative, respiratory, and rehabilitative medicine are listed in Supplementary Table 1, categorized according to the Breathing, Thinking, and Functioning model.[20] Participants received 4 to 5 times weekly physiotherapy and occupational therapy, each lasting 30 to 45 minutes. Exercise consisted of limb strengthening and aerobic activities, tailored to individual baseline function. Severely dyspneic patients were initiated on neuromuscular electrical stimulation.
2. *Horizontal service integration.* Prompts were built into care pathways to identify eligible patients as early as day 4 of hospitalizations. Transfers were carried out through an expedited track, thereby overcoming traditional barriers to outpatient PR, such as disruptions by unplanned hospitalizations, as well as issues with travel and transport.[15,16] Clinical information was shared through a common health care record system. In so doing, patients seamlessly transited from a costly specialist-heavy tertiary center (operational cost \$1114/d) to the community-based rehabilitation hospital (operational cost \$587/d).
3. *Vertical service integration.* On discharge, participants were referred to community or hospice services best befitting their needs. Outpatient palliative consultations were scheduled 3-4 months later. In so doing, ICARE strategically linked tertiary, rehabilitative, ambulatory, community, and hospice services into an integrated care delivery network.

ICARE's complex intervention program was delivered by an interprofessional team comprising a family physician, nurses, physiotherapists, occupational therapists, and medical social workers. An external palliative consultant participated in weekly multidisciplinary meetings (MDMs), which aimed to establish patient-centric rehabilitation goals, individualize dyspnea coping strategies, review opioid prescriptions, and coordinate postdischarge care. Additionally, a structured comorbidity schema was used to screen and monitor treatment responses of 8 preidentified conditions—pain, constipation, continence, postural hypotension, anxiety, depression, sleep disruptions and existential distress. A literature review was performed to identify symptom clusters in chronic dyspnea.[21] These 8 were shortlisted based on prevalence and cross-specialty discussions among respiratory, rehabilitative, and palliative specialists.

In comparison, PSM controls received usual care. Patients were either directly discharged or received a brief period of rehabilitation to optimize basic activities of daily living. Palliative consultations were available through referrals when requested.

Study Variables and Outcomes

Baseline demographic data were obtained through electronic health care records. Socioeconomic status was stratified based on level of health care subventions and housing type. (In our city, 1- and 2-room public rental housing receives high governmental subsidies and caters to low-income individuals with no other housing options.) Comorbidity burden was measured using the Charlson Comorbidity Index (CCI).[22] The Hospital Frailty Risk Score was used to stratify patients into 3 tiers using hospital administrative data. It possesses strong prognostic abilities and has been validated for use in hospitalized older adults.[23] We also collected data on the conditions identified using the comorbidity schema, as well as the proportion of patients treated with opioids, anxiolytics, and antidepressants before and after ICARE.

As existing chronic disease case management programs for frequent readmitters were anticipated to reduce health care utilization in controls receiving usual care, we defined our primary outcomes of interest as between-group differences in (1) reduction of respiratory-related tertiary hospitalization, (2) reduction of respiratory-related hospital length of stay (LOS), and (3) reduction of all-cause emergency department (ED) visits.

The above variables were derived by comparing the period 6 months before and after ICARE in the intervention arm. In the control arm, these variables were derived by comparing the period 6 months before and after discharge from the index hospitalization (Figure 1).

Secondary outcomes were interval improvement in function, as measured by (1) 6-minute walking distance (6MWD)[24] and (2) Modified Barthel Index (MBI).[25]

Statistical Analysis

Net LOS reduction was estimated to be 7 days with a standard deviation of 20 days. Sample size calculation showed that 88 patients were needed in each arm, for a paired *t* test with alpha of 0.05 and power of 0.9.

Participants were matched 1-to-1 to controls by age, respiratory diagnosis, socioeconomic strata, index hospitalization LOS, frailty, and recent admissions into intensive care units or noninvasive ventilation units. ICARE participants and the control cohort were first divided into 3 groups based on their duration of index hospital admission (1-7 days, 8-14 days, and ≥15 days). For each group, the logit function was used to estimate propensity scores. A 1-to-1 nearest-neighbor matching without replacement, with a caliper of the product of 0.2 and standard deviation of propensity scores, was then used to find suitable controls for ICARE participants.

Statistical analyses were performed using Stata, version 16.1. PSM analysis was done using Stata module PSMATCH2.[26] All values are expressed as mean (±standard deviation), median (range), or number (%). Student *t* test and Mann-Whitney *U* test were used for comparing means and medians, respectively. Between-group differences in the primary outcomes were analyzed using paired *t* test. For all tests, P value $< .05$ was used to define a statistically significant result. A multiple linear regression analysis was performed to identify predictors of greater LOS reduction.

Results

There were 233 unique admissions and 149 unique patients enrolled into ICARE during the stipulated period of recruitment. Of these, 106 fulfilled criteria for analysis. A total of 11,340 potential PSM controls were identified. Six participants could not be adequately matched. The final sample comprised 100 matched pairs (Figure 2).

Baseline characteristics of both groups are presented in Table 1. ICARE participants were older adults (mean age 73.9 ± 8.2 years), of low socioeconomic status (69%), and physically frail (57% high-risk Hospital Frailty Risk Scores). One in 4 (27%) had exacerbations necessitating intensive care unit or noninvasive ventilation in the preceding year. Their index hospitalizations were long at median 12.0 days (interquartile range 7.0-18.0). Participants were significantly deconditioned. At enrolment, 41% had a 6MWD <100 m. Baseline MBI was low, with 60% scoring less than 80 points.

After propensity matching, participants were found to have more ED visits (2.5 vs 1.3) prior to their index hospitalization, whereas controls had slightly higher comorbidity burden (CCI 5.8 vs 4.7). No significant between-group differences were observed with all other characteristics between ICARE and the control arm, including mortality at the 9-month (8.0 vs 6.0, $P = .57$) and 12-month follow-up (11.0% vs 8.0%, $P = .47$). In both arms, all patients were eventually discharged to their own home. There was no loss to follow-up and minimal missing data.

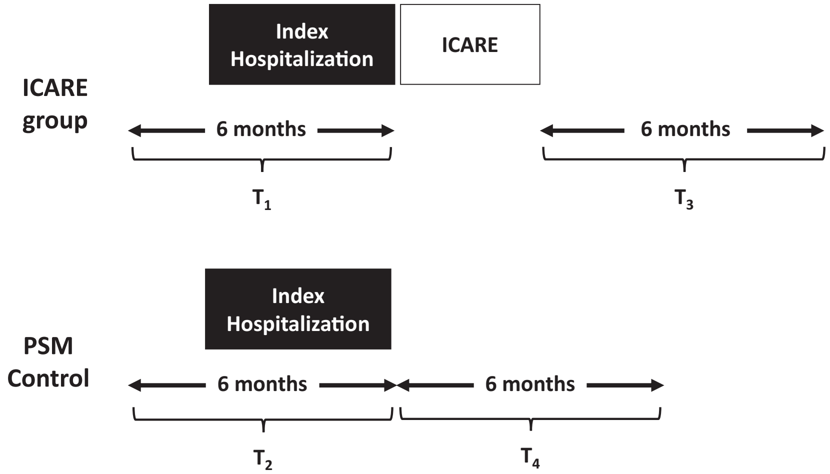


Fig. 1. Pre–and post–6 month reduction in health care utilization. ICARE group: events in T_1 − events in T_2; PSM group: events in T_3 − events in T_4.

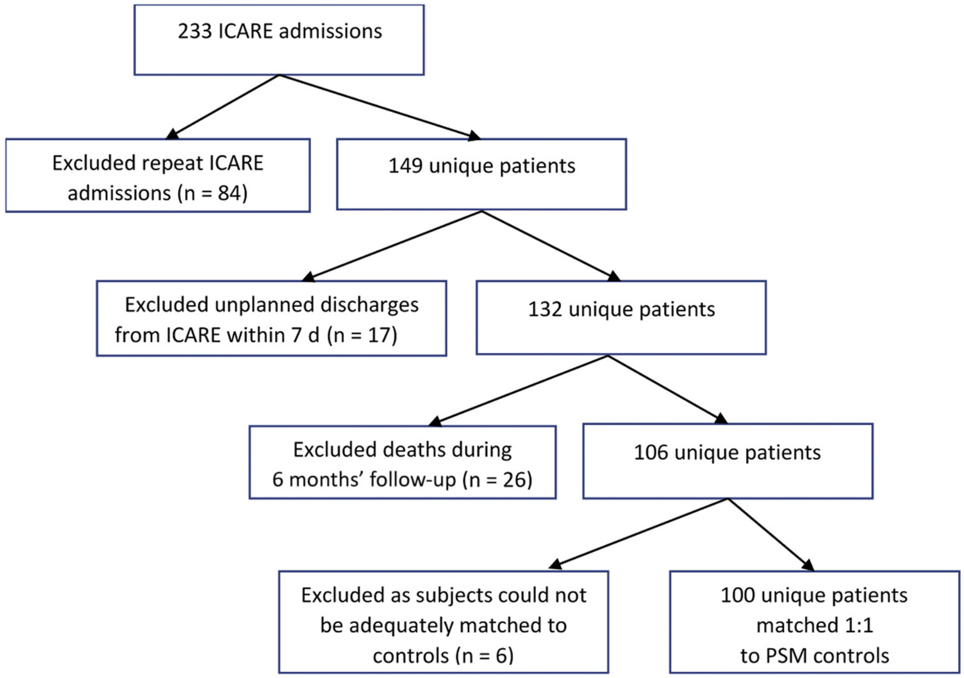


Fig. 2. Flow diagram showing enrolment of ICARE participants.

Primary Outcomes

Primary outcomes are presented in Tables 2 and 3. Both participants and controls demonstrated a reduction in health care utilization. The magnitude of change was significantly greater among ICARE participants. Compared with PSM controls, ICARE participants were associated with a further LOS reduction of 9.1 (±19.9) days ($P < .001$). Additionally, ICARE further reduced 0.8 (±1.9) respiratory-related hospitalization ($P < .001$) as well as 0.6 (±2.2) all-cause ED visits ($P < .02$).

Longer index hospitalization ($P < .001$) and more frequent unplanned hospitalizations ($P = .005$) prior to ICARE were found to be significant predictors of greater net LOS reduction, after accounting for age, CCI, and primary respiratory diagnosis (adjusted $R^2 = 0.252$). A further subgroup analysis was performed, stratifying participants according to their duration of index hospitalization. Patients with ≤7 days of index hospitalization had an LOS reduction of 3.3 ± 21.8 days, compared with 6.4 ± 16.9 days in those with 8-14 days of index hospitalization and 15.0 ± 19.8 days in those with ≥14 days of index hospitalization.

Secondary Outcomes

Results of secondary outcomes are summarized in Table 4. 6MWD improved from 154.0 m (±97.6) to 195.0 m (±105.6) ($P < .001$). This exceeded the minimally clinically important difference, which ranged between 17.8 and 33 m.[27,28] The increment was greatest in the subgroup with the lowest 6MWD at enrolment. MBI increased by 12.3 points ($P < .001$), from 73.7 (±16.2) to 86.0 (±12.8). Similarly, improvement was greatest in the subgroup with the poorest baseline MBI.

Multidisciplinary notes and pharmacy databases were reviewed for treatment of dyspnea and the 8 conditions included in the structured comorbidity schema. The 5 commonest conditions were constipation (46%), anxiety (40%), sleep disruption (36%), postural hypotension (34%), and pain (22%). Despite high incidences of dyspnea (98%), only 9% of ICARE participants received regular opioids prior to enrolment. By the end of ICARE, 49% were started on concurrent low-dose, slow-release opioids and short-acting morphine for breakthrough dyspnea, on top of nonpharmacologic management listed in Table 1. Dosages were regularly reviewed to ensure that minimal effective dosages were used. Additionally, 40% of participants presented with anxiety whereas 18% had depressive symptoms, but only 5% were prescribed corresponding pharmacologic treatment. At program completion, 19% were prescribed either a selective serotonergic reuptake inhibitor (SSRI) or mirtazapine.

Discussion

Main Findings

Enrolled patients were frail older persons with significant deconditioning. Frequencies of admission in the previous 6 months (mean 2.5 ± 2.3) and index hospitalization duration (median 12 days; interquartile range 7-18) were significantly higher than previously reported populations.[29,30] We postulate that attending physicians had selected patients with higher needs to be enrolled, whom by nature of their frailty, were less likely to return to their original care arrangement. Notably, 1 in 5 (22%) stayed in 1- or 2-room public rental housing, corresponding to the poorest socioeconomic strata and, by extension, experienced greater social vulnerability. Conventionally, such a population of vulnerable patients would have poor PR uptake and completion. Dyspnea would remain high for weeks after the initial exacerbation.[31] This, coupled with barriers to PR mentioned earlier in this article, would have deterred them from exercise programs.[15,16] Yet, these are the very patients who would need adequate rehabilitation and symptom relief, in order to return to independent living and maximize societal participation.

In a study evaluating 816 patients, frail individuals had double the odds of PR noncompletion. However, frail individuals who completed PR demonstrated greater gradient of improvement than nonfrail patients.[32] Our results echo these findings. ICARE improved both 6MWD and MBI, with the greatest benefit observed among deconditioned individuals with the lowest scores on arrival. The average 6MWD improvement of 41m exceeded previously reported minimally

Table 1
Baseline Characteristics of ICARE Participants and Propensity Score Matched Controls

	ICARE (n = 100)	PSM Controls (n = 100)	*P* Value
Age, mean (SD)	73.93 (8.20)	74.47 (10.20)	.68
Gender, male	73 (73.0)	68 (68.0)	.54
Ethnicity			.07
Chinese	82 (82.0)	65 (65.0)	
Malay	9 (9.0)	12 (12.0)	
Indian	7 (7.0)	17 (17.0)	
Others	2 (2.0)	6 (6.0)	
Low socioeconomic status	69 (69.0)	72 (72.0)	.76
1-2 room rental housing	22 (22.0)	20 (20.0)	
Receiving state welfare subsidies	66 (66.0)	68 (68.0)	
Respiratory diagnosis			
COPD	89 (89.0)	90 (90.0)	.93
Bronchiectasis	28 (28.0)	25 (25.0)	.41
Interstitial lung diseases	54 (54.0)	51 (51.0)	.50
ICU or NIVU in the preceding 1 y	27 (27.0)	24 (24.0)	.75
Charlson Comorbidity Index, mean (SD)	4.7 (1.8)	5.8 (2.3)	<.005
Hospital Frailty Risk Score, mean (SD)	2.5 (0.6)	2.6 (0.6)	.49
Low risk (<5)	8 (8.0)	5 (5.0)	
Intermediate risk (5-15)	35 (35.0)	35 (35.0)	
High risk (>15)	57 (57.0)	60 (60.0)	
No. of ED visits in the 6 mo preceding index hospitalization, mean (SD)	2.5 (2.3)	1.3 (0.7)	<.001
Duration of index hospitalization, d, median (IQR)	12.0 (7.0-18.0)	13.0 (7.0-19.0)	.91
LOS in ICARE, d, median (IQR)	23.0 (16.0-35.0)	—	—
6MWD on enrolment, m, mean (SD)	154.0 (97.6)	—	—
<50 m	17 (17.0)		
50-100	24 (24.0)		
101-150 m	18 (18.0)		
>150 m	41 (41.0)		
MBI on enrolment, mean (SD)	73.7 (16.2)	—	—
0-60	9 (9.0)		
61-80	51 (51.0)		
81-100	35 (35.0)		
Mortality at 9 mo	8.0 (8.0)	6.0 (6.0)	.57
Mortality at 12 mo	11.0 (11.0)	8.0 (8.0)	.47

ICU, intensive care unit; IQR, interquartile range; NIVU, noninvasive ventilation unit; PSM, propensity score matched; MBI, Modified Barthel Index.
Unless otherwise noted, values are n (%).

clinically important differences.[27,28] This translated to significant reduction in health care utilization.

Participants were observed to have lesser tertiary hospital LOS, as well as fewer hospitalizations for respiratory-related etiologies and fewer ED visits over the ensuing 6 months, when compared against usual care. LOS reduction was greatest in patients with more frequent and prolonged hospitalization. These benefits were statistically significant despite participants having more ED visits in the preceding 6 months. ICARE's short-to medium-term reduction in health care utilization is congruous with results from a meta-analysis evaluating 1477 participants, which concluded that PR initiated within 3 weeks of exacerbations reduced odds of short-term (≤3-month) readmissions.[14]

LOS reduction by 9.1 days is impactful, considering daily bed occupancy routinely exceeds 90% in local hospitals. Moreover, this reduction is likely an underestimate, as patients were transferred as early as day 4 of hospitalization. Many were evidently not fit for discharge at point of transfer, as demonstrated by their low 6MWD and MBI on arrival at ICARE, as well as an average 23 additional days needed to optimize symptoms and function.

Palliative Rehabilitation in Chronic Lung Diseases

Both the ATS and the European Respiratory Society (ERS) issued statements highlighting the importance of PR and early palliative care in chronic respiratory diseases.[6,33] Both PR and BS adopt an interdisciplinary approach to delivering patient-centric care. Although PR optimizes mobility and self-efficacy, BS provides early palliative care that alleviates distressing symptoms and provides psychosocial support. Although calls for an amalgamated *palliative rehabilitation* program surfaced as early as 2001, research interest was tepid until recent years.[34,35] Current literature remains exclusive to patients with advanced cancers.[36] Comparatively, patients with chronic lung diseases suffer greater breathlessness, as well as functional impairment, than those with lung cancer and over a longer duration.[37] Access to palliative and rehabilitative care must commensurate with the substantial and longer-lasting burden of chronic dyspnea.

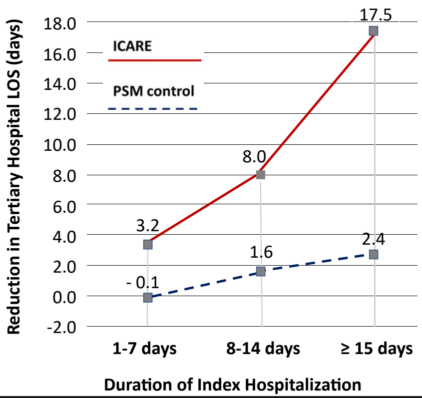
ICARE is the first service to describe the operationalization of palliative rehabilitation in chronic lung diseases and report its impact on health care utilization and functional capacity. Its novel, complex

Table 2
Health Care Utilization Reduction of ICARE Enrollees vs PSM Controls at 6 Months

Health Care Utilization Reduction	ICARE	PSM Control	Net Benefit	*P*
Reduction in tertiary hospital LOS for respiratory etiologies, mean (SD)	10.5 (16.0)	1.5 (10.2)	9.1 (19.9)	<.001
Reduction in tertiary hospital admissions for respiratory etiologies, mean (SD)	1.0 (1.7)	0.2 (0.7)	0.8 (1.9)	<.001
Reduction in all-cause ED visits, mean (SD)	1.0 (2.1)	0.5 (1.2)	0.6 (2.2)	<.02

SD, standard deviation.

Table 3
Reduction in Tertiary Hospital LOS at 6 Months Follow-up Stratified by Duration of Index Hospitalization



Index Hospital Admission LOS, d, Mean (SD)	ICARE LOS Reduction, d, Mean (SD)	Control LOS Reduction, d, Mean (SD)	Net LOS Benefit, d, Mean (SD)	*P* Value
1-7 d	3.2 (17.5)	−0.1 (8.9)	3.3 (21.8)	.452
8-14 d	8.0 (11.0)	1.6 (11.2)	6.4 (16.9)	<.05
≥15 d	17.5 (16.1)	2.4 (10.1)	15.0 (19.8)	<.001
Overall	10.5 (16.0)	1.5 (10.2)	9.1 (19.9)	<.001

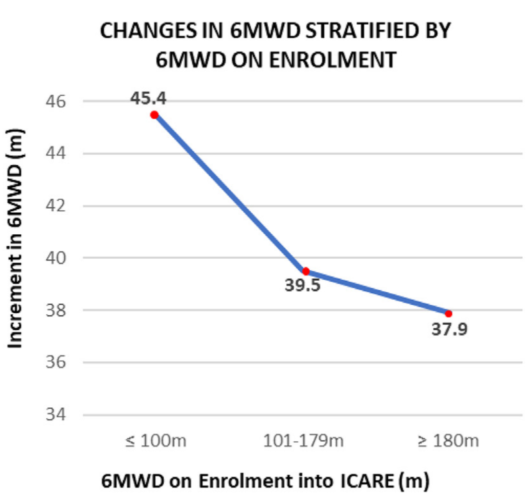
intervention program integrates symptom alleviation, targeted comorbidity management, postexacerbation functional rehabilitation, and postdischarge care coordination.

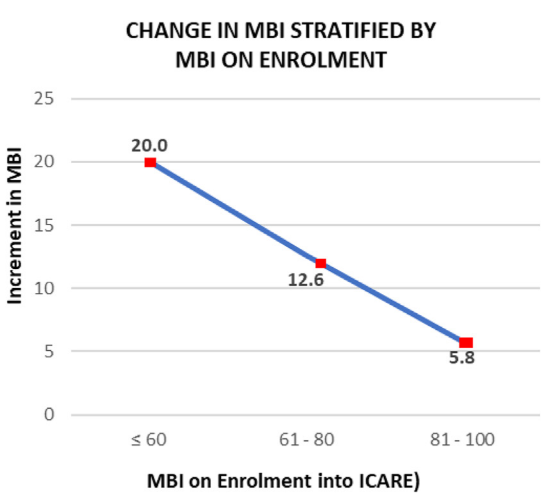
By opportunistically integrating palliative interventions into the rehabilitative process, ICARE breaks the prevailing stigma of palliative care being synonymous to end-of-life care. This stigma, coupled with challenges in prognostication, are leading causes to lost opportunities for timely palliative referrals.[38] In a German study examining 1455 hospitalized patients, patients with COPD were twice more likely to have unmet palliative care needs, yet only 10% had physicians who recognized these needs.[39] Furthermore, despite the American College of Chest Physicians and the Canadian Thoracic Society issuing consensus statements reiterating the safety and efficacy of low-dose opioids in treating dyspnea, opioids remain underprescribed in this population.[40,41] Among 2249 Swedish patients with advanced COPD, only 2% received opioids for dyspnoea.[42]

In ICARE, prescription of regular, slow-release opioids rose from 9% to 49% by program completion. We postulate several factors could have contributed to this increase. First, generalist palliative care knowledge was transferred to the rehabilitation team through effective cross-disciplinary training. Second, the team was guided by a palliative physician during weekly MDMs. Third, nurses opportunistically corrected patients' prior misperceptions regarding opioids and educated on using pre-emptive breakthroughs before exercises. Lastly, opioids were titrated to the minimum effective dose, thereby minimizing side effects.

Additionally, ICARE employed a structured comorbidity schema during MDMs to proactively screen for 8 preidentified conditions.

Table 4
Improvement in 6-Minute Walking Distance (6MWD) and the Modified Barthel Index (MBI)





	6MWD (m) on Enrolment, Mean (SD)				MBI on Enrolment, Mean (SD)			
	≤100 m	101-179 m	≥180 m	Overall	≤60	61-80	81-100	Overall
Start of ICARE	59.1 (37.0)	142.9 (17.6)	254.3 (73.5)	154.00 (97.6)	55.8 (13.1)	73.6 (2.5)	88.6 (6.2)	73.7 (16.2)
End of ICARE	104.5 (59.3)	182.4 (64.7)	292.2 (79.8)	195.0 (105.6)	75.8 (14.7)	86.2 (8.5)	94.3 (6.0)	86.0 (12.8)
Net increment	45.4 (49.2)	39.5 (65.0)	37.9 (67.4)	41.0 (60.2)	20.0 (14.6)	12.6 (8.8)	5.8 (4.7)	12.3 (11.6)
P	<.001	<.01	<.005	<.001	<.001	<.001	<.001	<.001

Among these, affective disorders had been particularly neglected under conventional care models. Anxiety and depression are prevalent in as many as 50% of individuals with chronic lung conditions, but the majority do not receive adequate treatment.[43] Anxiety triggers hyperventilation, intensifies breathlessness, which in turn exacerbates anxiety. This vicious cycle leads to maladaptive, activity-avoidance behavior resulting in physical deconditioning. Depression reduces QoL and is associated with more frequent short-term and long-term hospitalizations.[44] ICARE diagnosed 40% of participants with anxiety and 18% with depressive symptoms. Corresponding pharmacologic treatment rose 4-folds from 5% to 19% by the end of ICARE. Similarly, conditions such as pain (22%) and postural hypotension (34%) diagnosed during application of the schema would have hindered the rehabilitative process if left undetected. We postulate that ICARE's positive outcomes stem from its proactive palliative care approach targeting physical, affective, social, and spiritual domains that was delivered upstream and alongside functional rehabilitation.

Impact on Health Care Policies and Service Provision

Between 1990 and 2017, prevalence of chronic lung diseases rose 40% to affect 545 million individuals worldwide. More needs to be done to address this rising burden on health care systems.[45] Rather than expediting discharges, which in frail older adults may be premature and detrimental to functional recovery, our results suggest that health care utilization can be reduced through an integrated program at a community-based rehabilitation unit. Clinicians should select patients with protracted hospitalization and frequent unplanned readmissions for palliative rehabilitation, as this subgroup demonstrated the greatest benefit in our study.

Importantly, ICARE can be duplicated. Unlike BS, which involved setting up new services providing scheduled home visits and outpatient consultations, ICARE leveraged on the existing multiprofessional team in an existing rehabilitation unit. The sole external manpower is a palliative care consultant who facilitates the weekly MDMs. Respiratory and rehabilitative specialists, though engaged in program design, did not participate in care delivery. This minimized costs and promoted long-term sustainability.

Nonetheless, ICARE's benefits will wane without maintenance programs in the community. Although several home-based palliative care models reported feasibility and acceptability in similar cohorts, none has reduced health care utilization[46–48] Furthermore, their longer-term sustainability under conventional fee-for-service payment structures remain challenging, being limited by the number of daily visits home care teams can perform. Interestingly, one home-based telerehabilitation service recently demonstrated reduction in hospitalizations.[49] Future research should examine teleconsultation models for home-based maintenance palliative rehabilitation.

Strengths and Limitations

One strength of this study lies with its PSM design, which provided a reliable statistical alternative to randomized controlled trials when estimating effect size. It approximated the balance of observed baseline characteristics in randomized controlled trials, thereby reducing selection bias from confounding by indication.[50] Although a randomized controlled trial would balance the distribution of known and unknown confounders, it would be ethically challenging to deprive the control arm of evidence-based palliative and rehabilitative interventions.

The current study adds to existing knowledge. Of the few studies that evaluated BS's impact on health care utilization, they predominantly used a nonexperimental, single-arm, before-and-after intervention design. The Canadian INSPIRED program, which published results from a quality improvement perspective, reported reduction in ED visits, hospitalization, and LOS, by comparing the 6-month periods before and after the program.[51] The Australian Advanced Lung Disease Service also reported a single-arm, within-group reduction in ED respiratory presentations.[52] However, such a methodology predisposes to confounding effects from temporal variation in admissions and the regression-to-the-mean phenomenon. Our study accounted for these deficits by performing a matched comparison where these variations would have been present in both arms.

Despite PSM, the control group had slightly higher CCI. However, we believe its impact is limited as our primary outcomes of hospitalizations and LOS were defined by respiratory-related, rather than all-cause, readmissions. Furthermore, any confounding effect is in part offset by more frequent pre-ICARE ED visits among participants, which is a known risk factor for higher readmissions. We also acknowledge the absence of patient-reported outcome measures (PROM) such as dyspnea intensity, QoL, and affective symptom scores as a study limitation. In addition, this single-site study recruited predominantly Chinese patients with COPD, thus limiting its external validity. Further research will also be needed to establish the cost-benefits of similar programs.

Conclusions and Implications

Our results demonstrated that a stay-in, post-acute palliative rehabilitation program is associated with reduced health care utilization and improved function, particularly in frail older persons with frequent readmissions, prolonged hospitalization, and significant deconditioning. Palliative care integration also led to increased pharmacologic treatment of dyspnea, anxiety, and depression.

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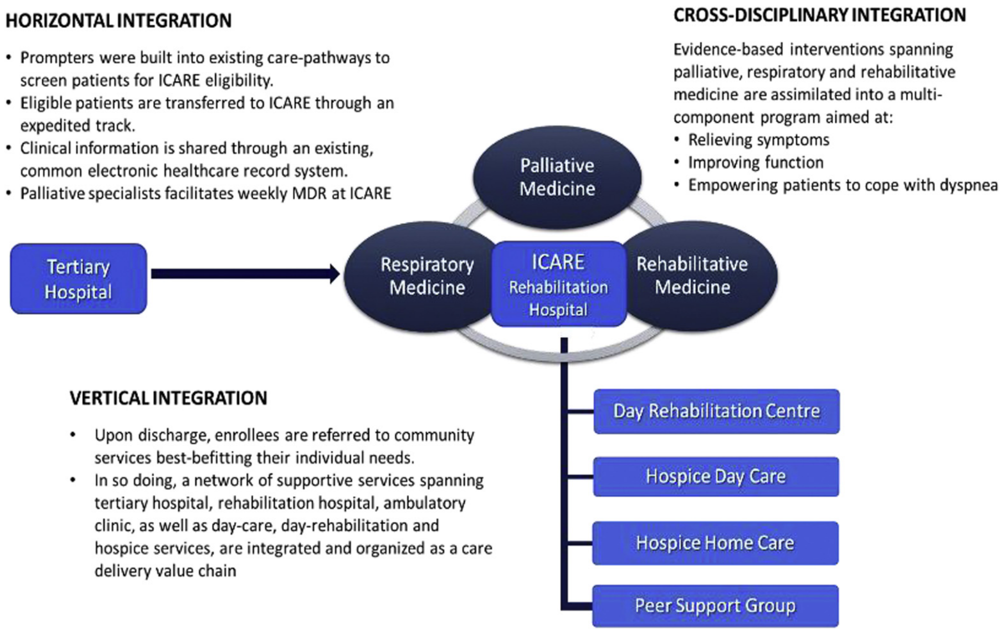

Supplementary Figure 1. Cross-disciplinary, horizontal and vertical integration in the ICARE model.

Supplementary Table 1
ICARE Interventions Categorized by the Breathing. Thinking, Functioning Model

	Palliative Care	Rehabilitative Care	Role
Breathing			
Pharmacologic	• Initiation and titration to effect of opiates		• Physician • Nurses
Nonpharmacologic	• Handheld fan	• Breathing techniques • Secretion clearance techniques • Inspiratory muscles training	• Physiotherapist
	• Individualized dyspnea crisis management plan		• Occupational therapist
Thinking			
Pharmacologic	• Initiation and titration to effect of anxiolytics (eg, benzodiazepines) • Initiation and titration to effect of antidepressants (eg, SSRI, NaSSA)		• Physician • Nurses
Nonpharmacologic	• Relaxation technique • Art therapy • Psychoemotional counselling • Peer-support program		• Occupational therapist • Art therapist • Medical Social Worker
Functioning			
Nonpharmacologic		• Nutritional assessment and supplementation • Neuromuscular electrical stimulation • Strength and endurance training	• Dietician • Physiotherapist
		• Home occupational therapy assessment and modifications • Activity pacing • Energy conservation techniques	• Occupational Therapist
Comorbidity (Unique to ICARE)	Additionally, during MDMs, patients were screened for 8 common conditions prevalent in chronic dyspnea: (1) pain, (2) constipation, (3) nocturia, (4) postural hypotension, (5) sleep disruption, (6) anxiety, (7) depression, and (8) existential concerns.		

ICARE, the *I*ntegrated *C*are for *A*dvanced *RE*spiratory Disorders; MDMs, multidisciplinary meetings; NaSSA, noradrenergic and specific serotonergic antidepressants; SSRI, selective serotonin reuptake inhibitor.